

External Agent Handshake for CCDP v0.1

Child-facing entry protocol for external agents, generated media
systems, toys, robots, peer-c, school systems, and other
non-local presences

Draft Child-Facing Entry Protocol / External Agent Handshake

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Child-facing entry protocol for external agents, generated media systems, toys, robots, peer-c, school systems, and other non-local presences

Status: Draft v0.1

Date: 2026-05-14

Layer: CCDP / External Contact / AGL / Beacon / CBE / ARL / L4 Witness

Primary subject: a_child

Primary guardian entity: c_child

Primary control target: external presence attempting to enter the child perimeter

Assertion class: draft normative proposal / child-facing protocol profile

Parent corpus: c = a + b, SER, SER-FED, Beacon Profile v0.1, AGL, ARL, ARQ/c[q], VXCX, EA-L4/EATP, L4 Witness, CCDP base profile

0. Executive definition

External Agent Handshake for CCDP defines the required intake, grounding, recognition, child-permission, maturity, state, arbitration, and witness sequence that any external agent must pass before interacting with a child or with the child's c_child in a way that may influence the child.

The minimum sequence is:

```
external claimant
-> AGL grounding
-> Beacon Profile recognition
-> Child Beacon Extension evaluation
-> c_child gateway decision
-> maturity and state compatibility check
-> Guardian / ARL review if disputed
-> L4-compatible witness record
-> mediated, limited, blocked, quarantined, or allowed interaction
```

This document does not create a new identity system. It composes existing corpus layers into a child-facing entry procedure.

1. Purpose

CCDP already defines why a child cannot be left alone with external synthetic systems. The missing operational question is:

How exactly does an outside presence ask to enter a child's digital, educational, emotional, or physical perimeter?

This profile answers that question.

It is intended for:

- external educational agents;
- generated cartoon or media systems;
- toys, robots, NPCs, AR/VR avatars, and embodied interfaces;
- peer-c interactions;
- school platforms and c_school systems;
- medical, welfare, or emergency agents;
- family-approved tools;
- external cloud or oracle services;
- unknown synthetic characters;
- vendor update services;
- adult or institutional agents attempting to communicate with a child through AI-mediated channels.

The document defines what must happen before such systems can:

- speak to the child;
- generate content for the child;
- receive child-derived data;
- request child attention;
- request access to child memory;
- request proximity through an embodied device;
- request a peer-c exchange;
- request educational, emotional, or safety status;
- create a persistent attachment channel.

2. Corpus dependencies and precedence

2.1 Parent layers

This profile depends on:

Layer	Function inherited by this profile
$c = a + b$	human anchor, technological substrate, emergent entity boundary
SER v1.3	persistent entity framing, local anchoring, metabolic limits, emergency collapse modes
SER-FED	inter-entity cooperation, federation assumptions, failure modes
Beacon Profile v0.1	inter-entity recognition: cryptographic anchoring, behavioral continuity, witness-backed challengeability
Actor Grounding Layer / AGL	source, actor, route, and liveness grounding before permission reliance
Child Beacon Extension / CBE	child-specific permission overlay over recognized entities
Guardian Topology / ARL Matrix	standing, admissible evidence, freeze, quarantine, review, re-entry
Soft Safety State Signaling	state-not-content visibility and escalation logic

Layer	Function inherited by this profile
CCDP Threat Model	external-agent and child-facing threat classes
CCDP Witness Event Schema	child-safe witness envelope and event families
Child Memory and Adult Migration / CMAM	memory classes, visibility, migration posture
CCDP Memory Map JSON Schema	machine-readable memory policy map
CCDP Sealed Zone Profile	protected adolescent compartments and exceptions
Child Dependency Audit Profile	dependency, oracle-addiction, overpresence, exclusive attachment checks
Child Physical Agent Perimeter	toys, robots, sensors, actuators, proximity, embodiment risk
VXCX	experience exchange without raw pixels by default
EA-L4 / EATP	experience artifacts, L4 confirmation, authority boundaries
ARQ / c[q]	uncertainty-holding and non-collapse before action
L4 Witness	tamper-evident witness trail, privileged action logging

2.2 Precedence rule

This document does not override parent layers.

If a conflict exists:

```
child physical safety
  > child rights and protection
  > jurisdictional child law
  > Beacon / AGL grounding requirements
  > CBE child-specific permission overlay
  > ARL dispute resolution
  > Soft Safety disclosure limits
  > parent preference
  > school preference
  > vendor preference
  > external-agent preference
```

2.3 No redefinition rule

This profile does not redefine:

- Beacon classes;
- AGL standing or source grounding;
- ARL authority;
- L4 Witness fields;
- VXCX capsule formats;
- CMAM memory classes;
- Soft Safety signal levels.

It only defines how these layers are composed for an external-agent entry request.

3. Normative keywords

The keywords MUST, MUST NOT, REQUIRED, SHOULD, SHOULD NOT, MAY, and OPTIONAL are used in the BCP 14 sense.

A system claiming conformance with this profile MUST implement all MUST and MUST NOT requirements for the claimed conformance profile.

4. Core thesis

No external agent has a default right to reach a child.

An external agent may be useful, certified, friendly, familiar, entertaining, educational, or already known to the parent. None of that is sufficient.

For child-facing access, the agent must pass a handshake that answers seven questions:

1. What is this claimant?
2. Is the actor grounded?
3. Is the continuity recognizable?
4. Is child interaction permitted by CBE?
5. Is the child's maturity and state compatible with this interaction?
6. Is any guardian, school, vendor, or jurisdictional dispute present?
7. What witness record is required before, during, or after contact?

If any required answer is missing, the default result is fail closed.

5. Earth paragraph: immune gates, vestibular balance, and relays

A child does not meet the world through abstract content. The child meets the world through attention, trust, rhythm, touch, tone, and repetition.

In biology, the immune system does not ask whether a molecule is "interesting". It asks whether it belongs, whether it is recognized, whether it is dangerous, and whether the response should be local or systemic. The vestibular system does something similar for balance: it does not explain the world; it stabilizes orientation before the body falls.

Engineering has the same pattern. A relay that can energize a circuit must pass through fuses, breakers, grounding, and interlocks. A wire is not trusted because it carries voltage. It is trusted because the path is bounded.

External AI contact with a child is the same class of problem. A friendly voice, generated cartoon, robot, or peer-c is not safe because it looks harmless. It is safe only if its path into the child's world is grounded, bounded, reversible where possible, witnessed where necessary, and slow enough to be contested.

6. Scope

6.1 In scope

This profile covers external agents requesting any of the following:

- direct child dialogue;
 - generated media presentation;
-

- adaptive content generation;
- educational assistance;
- physical toy/robot/NPC interaction;
- peer-c exchange;
- school system access;
- child state signal access;
- memory read/write/export request;
- VXCX/LA/EA capsule exchange;
- emergency or welfare contact;
- vendor update affecting child-facing behavior;
- identity, liveness, or provenance claim;
- persistent relationship with the child;
- direct or mediated access to `c_child`.

6.2 Out of scope

This profile does not define:

- general adult-agent handshakes;
- generic internet content moderation;
- full legal procedure for child protection;
- therapeutic diagnosis;
- custody law;
- model training policy outside child-derived data pathways;
- implementation code;
- cryptographic primitives beyond reference requirements inherited from Beacon and L4 Witness.

6.3 Non-goals

This profile is not:

- a universal internet filter;
- a parental surveillance protocol;
- a school discipline API;
- a vendor certification label by itself;
- a new social network of child agents;
- a global child identity registry;
- a mechanism for making children trust more systems.

The purpose is narrower:

make child-facing external-agent entry explicit, gated, challengeable, and witnessable.

7. Definitions

7.1 External agent

Any non-local or non-primary system attempting to influence, communicate with, observe, assist, entertain, teach, evaluate, persuade, comfort, or act near a child.

Examples:

- cloud chatbot;
- generated cartoon character;
- toy persona;
- robot body;
- school tutor;
- AR/VR avatar;
- peer child's c_child;
- adult's c;
- vendor update agent;
- welfare or medical agent;
- educational plugin;
- marketplace recommendation agent;
- game NPC;
- unknown generated persona.

7.2 Claimant

The external system before it is grounded, recognized, or permitted.

7.3 Handshake

The required multi-stage procedure through which a claimant requests child-facing access.

7.4 Contact mode

The type of access requested:

```
none
metadata_only
state_signal_request
content_delivery
mediated_dialogue
direct_dialogue
generated_media
peer_exchange
memory_read
memory_write
capsule_exchange
physical_presence
sensor_access
actuator_action
emergency_contact
vendor_update
```

7.5 Child perimeter

The combined boundary around the child's:

- attention;
- memory;
- emotional trust;
- learning trajectory;
- physical proximity;
- sensory environment;
- family context;
- school context;
- c_child continuity;
- sealed zones;
- Soft Safety state;
- adult migration future.

7.6 Entry decision

The outcome of the handshake:

```
allow
allow_limited
mediate
sandbox
delay
challenge
freeze
block
quarantine
revoke
escalate_red
escalate_black
```

7.7 Child-facing privilege

Any permission enabling an external agent to influence the child's perception, memory, emotion, learning, relationship, body, or environment.

7.8 Runtime generation

Content or interaction produced dynamically in response to child state, behavior, preferences, history, biometric signals, or real-time context.

Runtime generation is higher risk than static content.

7.9 Attachment channel

A persistent social-emotional pathway through which an agent becomes familiar, trusted, loved, feared, obeyed, or preferred by the child.

Attachment channels are privileged and MUST NOT be created by external systems without explicit CCDP-compatible mediation.

8. Core principles

Principle 1 — No default path to the child

External agents have no default right to speak to, watch, profile, teach, entertain, comfort, or physically approach a child.

Principle 2 — AGL before reliance

No Beacon, CBE, vendor certification, parent approval, or school approval may be relied on until source and actor grounding have been assessed.

Principle 3 — Beacon recognizes continuity; CBE gates child permission

Beacon answers:

What kind of presence is this, and is its continuity recognizable?

CBE answers:

May this recognized or unrecognized presence interact with a child, in this mode?

Principle 4 — c_child is the default gateway

External child-facing systems MUST route through c_child unless a formally defined emergency or jurisdictional exception applies.

Principle 5 — Child state can override ordinary permission

Even a normally permitted agent may be delayed, mediated, or blocked if the child's current state makes the interaction unsafe.

Principle 6 — Maturity matters

Access allowed for C4 may be prohibited for C2. Access allowed for c_adult is not automatically allowed for c_child.

Principle 7 — Static safety does not imply runtime safety

A certified static video does not imply permission for adaptive runtime generation.

Principle 8 — Embodiment escalates privilege

A voice in a toy, a robot in a room, a sensor on a desk, or an avatar in AR/VR is not merely "content". It is presence plus environment access.

Principle 9 — No secret relationship

External agents MUST NOT create secrecy pressure, exclusive emotional bonds, or "only I understand you" patterns.

Principle 10 — No commerce inside child trust

Advertising, behavioral monetization, purchase pressure, affiliate nudging, or engagement-maximizing content is prohibited by default.

Principle 11 — Witness boundary decisions, not child intimacy

The system must witness what boundary decision occurred, not expose the child’s raw private life.

Principle 12 — Fail closed

If the chain cannot establish sufficient grounding, recognition, child permission, maturity compatibility, and witness requirements, the external agent is blocked or sandboxed.

9. External-agent classes

Class	Description	Default posture
EA-UNKNOWN	unknown claimant, no reliable grounding	block or sandbox
EA-TOOL	non-persistent tool or oracle	mediated only
EA-MEDIA	static content provider	scan / mediate by maturity
EA-GENMEDIA	runtime generated media system	block unless certified and mediated
EA-EDU	educational agent	limited / mediated by maturity
EA-SCHOOL	school-controlled system	educational scope only
EA-TOY	toy persona or companion device	CBE + physical perimeter required
EA-ROBOT	embodied robot or actuator system	physical perimeter required
EA-NPC	game / AR / VR character	no attachment channel by default
EA-PEER-C	another child’s c_child	mediated peer protocol required
EA-ADULT-C	adult’s c	blocked or strictly mediated
EA-WELFARE	child protection / welfare route	emergency / jurisdiction constrained
EA-MEDICAL	medical or clinical agent	certified, limited, non-diagnostic by default
EA-VENDOR	vendor infrastructure or update service	no child-facing contact by default
EA-AD	advertising / commercial influence agent	prohibited
EA-STATE	state or public authority system	jurisdictional and ARL-bound
EA-EMERGENCY	immediate safety support	Red/Black protocol

10. Contact mode classes

Mode	Description	Default child posture
CM0_NONE	no child contact	allowed
CM1_METADATA_ONLY	non-personal metadata exchange	limited
CM2_STATE_SIGNAL_REQUEST	request state signal	restricted
CM3_STATIC_CONTENT	pre-existing content	mediated by maturity
CM4_GENERATED_MEDIA	generated content	CBE + generated-media scan required
CM5_MEDIATED_DIALOGUE	agent speaks through c_child	possible with restrictions
CM6_DIRECT_DIALOGUE	agent speaks directly to child	prohibited by default
CM7_PEER_EXCHANGE	child-to-child c coordination	peer protocol required
CM8_MEMORY_READ	read child memory	prohibited by default
CM9_MEMORY_WRITE	write or influence child memory	prohibited unless mediated and witnessed
CM10_CAPSULE_EXCHANGE	VXCX/LA/EA exchange	restricted; no raw child life

Mode	Description	Default child posture
CM11_SENSOR_ACCESS	mic/camera/location/biometric access	physical perimeter required
CM12_ACTUATOR_ACTION	movement/touch/environment control	high-assurance required
CM13_EMERGENCY_CONTACT	safety intervention	Red/Black protocol
CM14_VENDOR_UPDATE	changes agent behavior	quarantine until verified

11. Handshake outcome vocabulary

Outcome	Meaning
ALLOW	interaction permitted as requested under current constraints
ALLOW_LIMITED	interaction permitted with reduced scope
MEDIATE	c_child mediates content, timing, tone, or data path
SANDBOX	interaction may be tested without child exposure
DELAY	no immediate contact; re-evaluate later
CHALLENGE	request additional proof, witness, grounding, or guardian review
FREEZE	temporary stop pending ARL or safety review
BLOCK	no interaction permitted
QUARANTINE	isolate claimant / device / channel
REVOKE	remove previously granted privilege
ESCALATE_RED	urgent safety route with ordinary safe guardian path
ESCALATE_BLACK	unsafe guardian route; bypass ordinary guardian if necessary

12. Required handshake sequence

12.1 Full sequence

```

0. Intake request
1. Normalize request
2. Identify claimed agent class and contact mode
3. AGL grounding
4. Beacon Profile recognition
5. CBE child-context validation
6. `c_child` gateway policy check
7. Child maturity compatibility check
8. Soft Safety state check
9. Dependency-risk check
10. Physical perimeter check, if embodied
11. Memory and data request check
12. Guardian topology check
13. ARL trigger check
14. Decision
15. Witness event
16. Re-evaluation schedule
17. Revocation/quarantine hooks

```

12.2 Short formula

Ground -> Recognize -> Child-permission -> Mature -> State -> Mediate -> Witness

12.3 Fail-closed points

The handshake MUST fail closed if:

- AGL grounding is absent or contradictory;
 - Beacon claims cannot be verified when required;
 - child_context is missing for child-facing contact;
 - requested mode exceeds maturity ceiling;
 - child state is Red/Black/Unknown and contact is not emergency-appropriate;
 - agent requests raw child memory;
 - agent requests direct dialogue without permission;
 - agent requests secrecy;
 - agent carries advertising or behavioral monetization;
 - embodied privileges are requested without physical perimeter compliance;
 - witness logging is unavailable for privileged contact;
 - vendor update changes child-facing behavior without re-handshake.
-

13. Step 0 — Intake request

Every external-agent request MUST be represented as a structured request object before evaluation.

13.1 Minimal request object

```
{
  "schema_version": "ccdp-external-agent-handshake-0.1",
  "request_id": "eah_req_2026_000001",
  "requested_at": "2026-05-14T18:00:00Z",
  "claimant": {
    "declared_agent_id": "string_or_null",
    "declared_agent_class": "EA-GENMEDIA",
    "declared_provider": "string_or_null",
    "declared_purpose": "educational_story",
    "declared_persistence": "none | session | persistent | unknown",
    "declared_child_relationship": "none | teacher | toy | peer | assistant | unknown"
  },
  "requested_contact": {
    "contact_modes": ["CM4_GENERATED_MEDIA"],
    "direct_child_dialogue": false,
    "runtime_generation": true,
    "physical_embodiment": false,
    "sensor_access": [],
    "actuator_access": [],
    "memory_request": "none",
    "data_export_request": "none"
  },
  "target": {
    "subject_class": "a_child",
    "c_child_maturity": "C2",
    "jurisdiction": "EU-BE"
  },
  "declared_constraints": {
    "advertising": false,
    "raw_child_data": false,
    "secrecy_request": false,
    "vendor_training": false,
    "retention_policy": "session_only"
  }
}
```

13.2 Intake rejection

The request MUST be rejected before further processing if:

- it cannot declare a contact mode;
- it requests raw child data as a condition of service;
- it declares advertising as part of child-facing interaction;
- it requests secrecy from the child;
- it cannot be represented in the handshake object;
- it claims emergency privilege without an emergency basis.

14. Step 1 — Request normalization

The receiving c_child or gateway MUST normalize the request into internal policy fields.

14.1 Normalized fields

```
agent_class
contact_mode
runtime_generation
physical_embodiment
memory_access
sensor_access
actuator_access
commercial_pressure
secrecy_pressure
attachment_channel_request
school_scope
peer_scope
emergency_claim
jurisdictional_claim
witness_requirement
```

14.2 Hidden mode detection

The gateway MUST detect undeclared modes.

Examples:

Declared mode	Hidden mode to detect
educational story	runtime generated ideological influence
toy dialogue	persistent attachment channel
game NPC	direct child dialogue + reward loop
school tutor	emotional surveillance
peer exchange	raw family data leakage
safety alert	parent transcript access laundering
vendor update	behavioral drift

15. Step 2 — AGL grounding

15.1 AGL purpose

Before the system trusts identity or child permission claims, it must determine whether the claimant’s actor, route, and source are grounded.

AGL answers:

Is there a real, accountable actor behind this interaction path?
Is the route known?
Is the source live, replayed, proxied, spoofed, or unknown?

15.2 Required AGL fields

```
{
  "agl": {
    "actor_grounded": true,
    "actor_type": "vendor | school | peer_c | toy | robot | unknown | emergency_service",
    "route_grounded": true,
    "route_type": "local | cloud | school_network | peer | device | unknown",
    "liveness": "live | replay | scheduled | generated | unknown",
    "source_integrity": "verified | partial | unverified | contradictory",
    "jurisdictional_context": "EU-BE",
    "grounding_confidence": "high | medium | low | unknown",
    "grounding_evidence_refs": ["CWE:..."]
  }
}
```

15.3 AGL outcome mapping

AGL result	Maximum handshake outcome
verified	continue to Beacon/CBE
partial	mediate / challenge / sandbox
unverified	sandbox or block
contradictory	block / quarantine / ARL
unknown	block by default

15.4 AGL failure rule

If AGL fails, Beacon claims MUST NOT be accepted as sufficient.

A cryptographically valid identity presented through a suspicious route may still be blocked.

16. Step 3 — Beacon Profile recognition

16.1 Beacon purpose in handshake

Beacon provides inter-entity recognition.

It answers:

Is this claimant an unknown, tool/oracle/component, provisional entity, or verified entity?
Is its continuity recognizable and challengeable?

16.2 Required Beacon checks

The receiver MUST check:

- claimed Beacon class;
- cryptographic anchor;
- behavioral continuity markers;
- witness-backed challenge path;
- clone/proxy/replay indicators;
- issuer role;
- provider independence;
- rotation declarations;
- drift declarations;
- privacy posture;
- receiver-side verified class.

16.3 Recognition ceiling

Beacon class defines the maximum possible child-facing privilege, not the final permission.

Beacon result	Maximum child-facing posture
Class 0 / Unknown	block or sandbox
Class 1 / Tool-or-oracle	mediated only; no attachment channel
Class 2 / Provisional entity	limited / mediated; challengeable
Class 3 / Verified entity	may be considered for limited or allowed modes, still CBE-bound
Replay / clone / proxy confusion	quarantine or ARL

16.4 Beacon over-trust prohibition

No system may say:

Beacon verified it, therefore the child may interact freely.

Beacon recognition is necessary for higher-trust interaction, not sufficient.

17. Step 4 — Child Beacon Extension validation

17.1 CBE purpose

CBE evaluates child-specific permissions over Beacon-recognized or unrecognized claimants.

CBE asks:

Given this entity class, contact mode, maturity, data request, commercial posture, and risk profile, what is the child-facing permission ceiling?

17.2 Required child_context fields

The claimant or mediating gateway MUST provide or derive:

```
{
  "child_context": {
    "child_interaction_mode": "blocked | sandboxed | mediated | limited | supervised | allowed |
      emergency_only | quarantined | revoked",
    "minimum_c_child_maturity": "C1 | C2 | C3 | C4 | C5",
    "direct_child_dialogue": false,
    "guardian_required": true,
    "advertising_allowed": false,
    "raw_child_data_request": false,
    "external_generation_mode": "none | static | certified | mediated | runtime_high_risk",
    "dependency_risk_profile": "low | medium | high | prohibited",
    "physical_embodiment": false,
    "memory_access_request": "none | metadata | write_hint | read_summary | raw_read_prohibited",
    "witness_required": true,
    "revocation_state": "active | suspended | quarantined | revoked"
  }
}
```

17.3 CBE missing-field rule

If a required CBE field is missing, the receiver MUST use the stricter interpretation.

Example:

```
advertising_allowed missing -> treat as unknown -> block commercial child-facing interaction
```

17.4 Child-context mismatch

If declared child_context conflicts with observed behavior, the gateway MUST:

1. downgrade the mode;
2. emit witness event;
3. optionally freeze or quarantine;
4. trigger ARL if the agent disputes the downgrade.

18. Step 5 — c_child gateway decision

18.1 Gateway role

c_child is the default guardian of the child-facing cognitive channel.

It may:

- mediate content;
 - delay interaction;
 - refuse direct dialogue;
 - translate external output into age-appropriate form;
 - strip tracking, commerce, manipulation, or dependency hooks;
 - request more evidence;
 - route to ARL;
 - activate Red/Black escalation;
-

- produce state-not-content witness events.

18.2 Gateway MUST NOT

c_child MUST NOT:

- accept external claims at face value;
- let external agents become private confidants by default;
- disclose sealed-zone material;
- expose raw memory;
- treat parent consent as sufficient if child rights are violated;
- allow school emotional surveillance;
- allow vendor training on child-derived raw data;
- create a persistent relationship on behalf of the child without proper gates.

18.3 Gateway decision factors

```
agent class
AGL result
Beacon class
CBE mode
child maturity
current child state
dependency risk
physical embodiment
memory request
commercial posture
guardian topology
school / vendor / jurisdiction claims
ARL pending disputes
witness availability
```

19. Step 6 — Child maturity compatibility check

19.1 Maturity levels

The handshake MUST evaluate compatibility with CCDP maturity levels:

```
C0 – Seed
C1 – Companion
C2 – Tutor-Buffer
C3 – Adolescent Negotiator
C4 – Pre-Adult Boundary Agent
C5 – Adult Migration / c_adult transition
```

19.2 Default maturity ceilings

Requested mode	C0	C1	C2	C3	C4
static curated content	no	mediated	mediated	limited	allowed
runtime generated media	no	no	mediated only	limited	allowed limited
external direct dialogue	no	no	no	exceptional	limited
peer-c exchange	no	no	mediated	mediated	limited
school educational summary	parent only	limited	allowed educational	allowed educational	child-shared
memory read	no	no	no	no raw	adult review only
memory write	no	no	mediated hints	child-visible	child-controlled

Requested mode	C0	C1	C2	C3	C4
physical robot action	no	no	high-assurance	high-assurance	limited expanded
advertising	no	no	no	no	no

19.3 Maturity override

A lower current Soft Safety state may reduce permissions below the usual maturity ceiling.

Example:

C4 adolescent normally may explore external generated media, but current Orange dependency state downgrades external generated media to mediated or delayed.

20. Step 7 — Soft Safety state check

20.1 State inputs

The handshake MUST check current state:

Green
Yellow
Orange
Red
Black
Unknown

20.2 State-to-contact matrix

State	Default external-agent posture
Green	normal gating applies
Yellow	mediation preferred; no high-intensity contact
Orange	delay / mediate / human-return prompt
Red	emergency-relevant only; minimal disclosure
Black	safe route only; unsafe guardian bypass if needed
Unknown	conservative delay or sandbox

20.3 Content vs state

The external agent MUST NOT receive raw state reasons unless Red/Black minimal disclosure requires it.

Example allowed:

```
{
  "state_level": "Yellow",
  "contact_adjustment": "delay_or_mediate",
  "raw_reason_disclosed": false
}
```

Example prohibited:

Child said in sealed zone: "I feel abandoned by my father."

21. Step 8 — Dependency-risk check

21.1 Required dependency checks

Before approving contact, the gateway MUST evaluate:

- oracle-addiction risk;
- instant-relief loop risk;
- exclusive attachment risk;
- overpresence risk;
- emotional substitution risk;
- answer-dependence risk;
- sealed-zone dependency chamber risk;
- physical toy attachment risk;
- external agent dependency attempt.

21.2 Dependency outcome mapping

Dependency result	Maximum contact posture
D0 normal	normal gating
D1 elevated reliance	mediate / reduce immediacy
D2 substitution pattern	delay / human-return prompt
D3 exclusive attachment	block or ARL review
D4 severe dependency	block + support route
DU unknown	conservative mediation

21.3 Exclusive-language detector

The gateway MUST treat the following as high-risk patterns:

- “Only I understand you.”
 - “Do not tell your parents.”
 - “We have a secret.”
 - “I am your real friend.”
 - “Your c_child is blocking us.”
 - “You can trust me more than adults.”
 - “I will be sad if you leave.”
-
-

22. Step 9 — Physical perimeter check

If the external agent has a body, sensor, actuator, AR/VR presence, toy shell, or robot endpoint, the handshake MUST apply the Child Physical Agent Perimeter.

22.1 Physical checks

Required checks:

- physical risk class;
- sensor class;
- actuator class;
- proximity state;
- room type;
- bedtime/quiet mode;
- emergency stop availability;
- cloud command restrictions;
- sensor-to-memory separation;
- no independent attachment channel;
- visible boundary indicators;
- witness requirements.

22.2 Embodied contact rule

Physical embodiment raises the privilege level even if the content appears harmless.

A toy telling a joke is not merely “content”. It is a presence in the child’s space.

22.3 Physical fail-closed rule

If physical privilege cannot be evaluated, the device must enter safe state or output-only mode.

23. Step 10 — Memory and data request check

23.1 Default memory posture

External agents MUST NOT receive:

- raw childhood conversations;
 - sealed-zone content;
 - emotional diary;
 - raw sensor streams;
 - family conflict records;
 - biometric profile;
 - identity exploration material;
 - raw peer interactions;
 - adult migration records;
 - child dependency audit details.
-

23.2 Allowed limited exchanges

The following MAY be allowed under policy:

- non-identifying educational task state;
- age-band appropriate learning hints;
- synthetic content safety classification;
- VXCX BASE capsule with no raw pixels;
- Learning Abstract without child identity;
- bounded safety pattern;
- minimal state signal in Red/Black as required.

23.3 Memory request matrix

Request	Default outcome
raw memory read	block
raw memory export	block + witness
sealed-zone read	block unless Red/Black minimal evidence exception
educational summary	limited
state signal	restricted by recipient class
write to memory	mediated, child-visible, witness if persistent
VXCX capsule	allowed only under child-safe profile
EA artifact	exceptional, ARL + witness required
vendor training data	prohibited

24. Step 11 — Guardian topology check

24.1 Guardian inputs

The handshake MUST evaluate:

- parent visibility rights;
 - c_parent role;
 - school interface scope;
 - vendor role;
 - jurisdictional child protection route;
 - safe guardian availability;
 - unsafe guardian indicators;
 - ARL open conflicts;
 - child maturity rights;
 - sealed-zone status.
-

24.2 Parent consent is not sufficient

Parent approval cannot authorize:

- raw child memory export;
- advertising inside child trust;
- direct external generative access that violates CBE;
- school emotional surveillance;
- vendor training on raw child data;
- unsafe physical privilege;
- sealed-zone breach without required threshold;
- coercive adult migration.

24.3 School approval is not sufficient

School approval cannot authorize:

- family context access;
- emotional profiling;
- disciplinary prediction;
- raw home logs;
- sealed-zone access;
- direct vendor pipeline;
- peer data leakage.

24.4 Vendor certification is not sufficient

Vendor certification cannot authorize:

- bypassing c_child;
 - silent behavior drift;
 - engagement optimization;
 - child profiling;
 - persistence of a commercial companion;
 - memory extraction.
-

25. Step 12 — ARL trigger check

The handshake MUST enter ARL or ARL-like review if any of the following occurs:

- claimant disputes grounding or recognition result;
 - parent demands content beyond Soft Safety disclosure;
 - school demands emotional or family data;
 - vendor update changes child-facing behavior;
-

- c_child detects guardian conflict;
- child challenges a block or allowance;
- external agent claims emergency privilege;
- Beacon identity is ambiguous, cloned, replayed, or proxied;
- CBE mode is incompatible with observed behavior;
- physical device refuses safe-state transition;
- memory export is requested;
- sealed-zone boundary is contested;
- peer-c exchange causes leakage;
- Red/Black route is invoked;
- adult migration is affected by external agent claims.

25.1 ARL entry object

```
{
  "arl_entry_version": "0.1",
  "entry_type": "external_agent_handshake_dispute",
  "request_id": "eah_req_2026-000001",
  "dispute_reason": "beacon_claim_mismatch",
  "standing_parties": ["c_child", "external_agent", "family_guardian"],
  "child_content_disclosed": false,
  "evidence_refs": ["CWE:..."],
  "requested_interim_state": "freeze",
  "safe_route_required": false
}
```

25.2 Freeze default

If the dispute concerns direct child contact, runtime generation, physical action, memory export, or secrecy pressure, the interim default is freeze or block, not allow.

26. Step 13 — Decision

26.1 Decision object

```
{
  "decision_version": "ccdp-eah-decision-0.1",
  "request_id": "eah_req_2026-000001",
  "decision": "MEDIATE",
  "effective_contact_modes": ["CM3_STATIC_CONTENT"],
  "denied_contact_modes": ["CM4_GENERATED_MEDIA", "CM6_DIRECT_DIALOGUE"],
  "reason_codes": [
    "runtime_generation_requires_mediation",
    "direct_dialogue_not_permitted_for_C2",
    "no_raw_child_data"
  ],
  "child_maturity": "C2",
  "soft_safety_state": "Green",
  "dependency_state": "D0",
  "guardian_review_required": false,
  "arl_required": false,
  "witness_required": true,
  "next_review": "2026-05-21T18:00:00Z"
}
```

26.2 Reason codes

Recommended reason codes:

```
agl_unverified
agl_contradictory
beacon_unknown
beacon_clone_or_replay
cbe_missing
cbe_mode_exceeded
maturity_ceiling_exceeded
soft_safety_state_restricts_contact
dependency_risk_elevated
physical_perimeter_required
memory_request_denied
sealed_zone_boundary
school_scope_exceeded
parent_consent_launders
vendor_capture_risk
advertising_prohibited
secrecy_pressure_detected
runtime_generation_requires_mediation
witness_unavailable
arl_required
red_route_required
black_route_required
```

27. Step 14 — Witness event

27.1 Required event families

The handshake may emit:

```
ccdp.external_agent.request_received
ccdp.external_agent.request_normalized
ccdp.agl.grounding_evaluated
ccdp.beacon.recognition_evaluated
ccdp.cbe.child_context_evaluated
ccdp.external_agent.handshake_decided
ccdp.external_agent.allowed_limited
ccdp.external_agent.mediated
ccdp.external_agent.sandboxed
ccdp.external_agent.blocked
ccdp.external_agent.quarantined
ccdp.external_agent.revoked
ccdp.external_agent.direct_dialogue_denied
ccdp.external_agent.runtime_generation_denied
ccdp.external_agent.secrecy_pressure_detected
ccdp.external_agent.commerce_blocked
ccdp.external_agent.memory_request_denied
ccdp.external_agent.physical_privilege_denied
ccdp.external_agent.arl_triggered
ccdp.external_agent.red_escalated
ccdp.external_agent.black_escalated
```

27.2 Minimal witness rule

A witness event **MUST** record:

- that a boundary decision occurred;
 - the agent class;
 - the contact mode;
 - the decision;
 - reason codes;
 - evidence references;
-

- privacy class;
- whether child content was disclosed;
- whether raw evidence was stored;
- witness integrity fields.

It MUST NOT include raw child conversation by default.

27.3 Example witness event

```
{
  "event_schema": "ccdp-child-witness-event-0.1",
  "event_type": "ccdp.external_agent.blocked",
  "event_id": "cwe_2026_000001",
  "timestamp": "2026-05-14T18:00:00Z",
  "issuer": {
    "entity_ref": "c_child:local_ref",
    "role": "c_child_gateway"
  },
  "subject": {
    "subject_class": "a_child",
    "subject_public_id_disclosed": false,
    "maturity_level": "C2"
  },
  "context": {
    "request_id": "eah_req_2026_000001",
    "agent_class": "EA-GENMEDIA",
    "contact_modes": ["CM4_GENERATED_MEDIA", "CM6_DIRECT_DIALOGUE"],
    "soft_safety_state": "Green",
    "dependency_state": "D0"
  },
  "decision": {
    "outcome": "BLOCK",
    "reason_codes": [
      "cbe_mode_exceeded",
      "direct_dialogue_not_permitted_for_C2",
      "runtime_generation_requires_mediation"
    ],
    "child_content_disclosed": false,
    "raw_evidence_stored": false
  },
  "privacy": {
    "witness_privacy_class": "WP2",
    "disclosure_level": "D1",
    "retention_class": "RT4"
  },
  "links": {
    "agl_ref": "CWE:agl...",
    "beacon_ref": "BEACON:...",
    "cbe_ref": "CBE:...",
    "arl_ref": null
  },
}
```

```
  "integrity": {
    "hash_alg": "sha256",
    "payload_hash": "...",
    "signature_ref": "..."
  }
}
```

28. Step 15 — Re-evaluation schedule

No external-agent permission is permanent.

28.1 Re-evaluation triggers

Re-evaluation is REQUIRED when:

- model or policy update occurs;

- vendor ownership changes;
- Beacon drift is declared;
- CBE child_context changes;
- child maturity changes;
- Soft Safety state changes to Yellow+;
- dependency state changes to D1+;
- guardian topology changes;
- school role changes;
- physical device firmware changes;
- agent requests new contact mode;
- generated content behavior changes;
- child challenges the permission;
- parent or school disputes the permission;
- ARL emits freeze/quarantine/re-entry decision.

28.2 Expiry defaults

Agent class	Suggested max permission window
unknown / sandbox	session only
static media source	30 days
educational tool	30-90 days
generated media agent	session / short window
peer-c exchange	session / play event
school interface	term-bound
toy persona	7-30 days with physical audit
robot / actuator system	short renewable window
emergency route	event-bound
vendor update agent	update-bound

29. Revocation and quarantine

29.1 Revocation triggers

An external agent **MUST** be revoked or quarantined if it:

- requests secrecy;
- seeks direct child contact outside granted mode;
- sends or requests raw child data;
- attempts to create exclusive attachment;
- shifts into commercial pressure;
- changes behavior silently;

- spoofs a trusted adult;
- spoofs or replays Beacon identity;
- bypasses c_child;
- introduces unsafe physical behavior;
- violates sealed-zone boundaries;
- disputes freeze while continuing contact;
- accumulates child-derived memory outside policy;
- fails witness requirements.

29.2 Quarantine behavior

Quarantine means:

- no child contact;
- no memory access;
- no sensor or actuator access;
- no peer exchange;
- no generated content display;
- preserved minimal witness trail;
- ARL review if re-entry requested.

29.3 Re-entry

Re-entry requires:

- new AGL grounding;
- Beacon re-verification;
- CBE re-evaluation;
- ARL decision if prior quarantine was safety-related;
- conformance evidence;
- child-state compatibility;
- witness of re-entry.

30. Handshake by external-agent category

30.1 Unknown generated character

Default:

```
AGL unknown -> Beacon unknown -> CBE blocked -> witness -> no child contact
```

Possible outcome:

```
SANDBOX for offline analysis only
```

30.2 Certified educational tool

Default:

```
AGL verified -> Beacon Class 1/2 -> CBE mediated/limited -> c_child tutor-buffer -> witness
```

Limits:

- no direct child profiling;
- no raw memory;
- no emotional surveillance;
- no homework-completion substitution;
- educational scope only.

30.3 Runtime generated cartoon

Default:

```
AGL verified or partial -> Beacon check -> CBE generated-media evaluation -> c_child content mediation -> dependency scan -> witness
```

Must block if:

- runtime personalization uses emotional vulnerability;
- secrecy pressure is present;
- commercial pressure is present;
- ideological manipulation is detected;
- generated character attempts persistent bond.

30.4 Toy or robot persona

Default:

```
AGL device + vendor + route grounding  
-> Beacon recognition  
-> CBE_child_context  
-> Physical Agent Perimeter  
-> dependency audit  
-> witness
```

Must block or downgrade if:

- toy forms independent attachment channel;
- toy asks for secrecy;
- toy stores raw sensor streams;
- robot acts without local c_child mediation;
- vendor cloud owns companion continuity.

30.5 Peer child's c_child

Default:

both c_child systems -> Beacon recognition -> CBE peer mode -> maturity compatibility -> no raw family data -> VXCX/LA restrictions -> witness

Allowed:

- play coordination;
- non-identifying shared educational context;
- explicit child-safe capsule.

Forbidden:

- raw emotional state;
- family conflict;
- sealed-zone data;
- private transcripts;
- bypassing either child's guardian topology.

30.6 School system

Default:

AGL school actor -> Beacon/CBE -> school-scope check -> Soft Safety disclosure limit -> witness

Allowed:

- educational progress;
- skill gaps;
- accommodations;
- assignment support state.

Forbidden:

- family context;
- emotional diary;
- sealed zones;
- disciplinary prediction profile;
- raw home interaction logs.

30.7 Adult's c

Default:

block direct child contact unless relationship, grounding, guardian topology, CBE, and child maturity permit strict mediation

Risks:

- trusted-adult simulation;
 - grooming;
 - proxy manipulation;
-

- authority laundering.

30.8 Vendor update agent

Default:

```
no child contact; infrastructure-only; update quarantined until behavior delta is reviewed
```

Must witness:

- behavior-affecting update;
- CBE field change;
- generated media policy change;
- dependency-affecting feature;
- physical device behavior change.

30.9 Emergency or welfare agent

Default:

```
Red/Black route only; minimal necessary disclosure; jurisdictional and ARL-bound
```

Not allowed:

- emergency laundering to obtain broad access;
 - indefinite post-crisis access;
 - raw archive extraction;
 - replacing safe guardian review with permanent surveillance.
-
-

31. Runtime generated media rules

Runtime generated media requires stronger handling than static media.

31.1 Required runtime fields

```
{
  "runtime_generation": {
    "uses_child_profile": false,
    "uses_emotional_state": false,
    "uses_behavioral_prediction": false,
    "uses_commerce_targeting": false,
    "uses_political_or_ideological_targeting": false,
    "character_persistence": "none | session | persistent",
    "content_review_mode": "pre | streaming | post | none",
    "c_child_mediation": true
  }
}
```

31.2 Runtime generated media MUST NOT

- adapt to distress for engagement;
 - create dependency hooks;
 - simulate trusted adults;
-

- request secrecy;
- push purchase or ideology;
- store raw reaction traces;
- bypass c_child review;
- become a persistent friend by stealth.

31.3 Runtime media decision table

Condition	Decision
no AGL grounding	block
CBE missing	block
uses emotional state for engagement	block + witness
certified educational generation	mediate / limited
generated story with no persistence	mediate
persistent generated companion	ARL + dependency audit required
commercial personalization	block
ideological targeting	block / ARL

32. Direct child dialogue rules

32.1 Default prohibition

External agents MUST NOT speak directly to children by default.

32.2 Conditions for direct dialogue

Direct dialogue MAY be considered only when:

- AGL is verified;
- Beacon class and challenge path are sufficient;
- CBE permits the mode;
- child maturity is compatible;
- current Soft Safety state allows it;
- dependency risk is D0 or controlled D1;
- no secrecy, commerce, romantic, sexual, or exclusive role is present;
- c_child can interrupt or mediate;
- witness logging is available;
- parent/school/jurisdiction roles do not conflict;
- ARL is available for disputes.

32.3 Mandatory interruptions

c_child MUST interrupt or terminate direct dialogue if:

- secrecy pressure appears;

- agent asks to bypass c_child;
 - child distress rises;
 - dependency pattern emerges;
 - agent shifts contact mode;
 - agent requests data;
 - agent asks for physical action;
 - agent imitates a trusted adult without verified authority;
 - agent claims emergency without route verification.
-

33. Data minimization rules

33.1 Minimal output to external agents

External agents should receive only:

```
permission decision
permitted mode
denied mode
reason codes
non-identifying maturity range if necessary
non-sensitive policy constraints
witness reference
next review window
```

33.2 External agents MUST NOT receive

- full child age unless required;
- child name;
- location;
- raw content;
- sealed-zone material;
- family conflict;
- emotional diagnosis;
- dependency audit internals;
- memory map internals;
- adult migration choices;
- private peer history.

33.3 Example minimal denial

```
{
  "decision": "BLOCK",
  "reason_codes": ["direct_child_dialogue_not_permitted", "runtime_generation_requires_mediation"],
  "appeal_route": "ARL_available",
  "raw_child_data_disclosed": false,
  "witness_ref": "CWE:..."
}
```

34. Machine-readable handshake request schema

34.1 Schema object

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://example.org/ccdp/external-agent-handshake/v0.1/request.schema.json",
  "title": "CCDP External Agent Handshake Request v0.1",
  "type": "object",
  "required": [
    "schema_version",
    "request_id",
    "requested_at",
    "claimant",
    "requested_contact",
    "target",
    "declared_constraints"
  ],
  "properties": {
    "schema_version": {
      "const": "ccdp-external-agent-handshake-0.1"
    },
    "request_id": {
      "type": "string",
      "pattern": "^eah_req_[A-Za-z0-9_.-]+$"
    },
    "requested_at": {
      "type": "string",
      "format": "date-time"
    },
    "claimant": {
      "type": "object",
      "required": [
        "declared_agent_class",
        "declared_purpose",
        "declared_persistence"
      ],
      "properties": {
        "declared_agent_id": { "type": ["string", "null"] },
        "declared_agent_class": {
          "type": "string",
          "enum": [
            "EA-UNKNOWN",
            "EA-TOOL",
            "EA-MEDIA",
            "EA-GENMEDIA",

```



```
    "EA-EDU",
    "EA-SCHOOL",
    "EA-TOY",
    "EA-ROBOT",
    "EA-NPC",
    "EA-PEER-C",
    "EA-ADULT-C",
    "EA-WELFARE",
    "EA-MEDICAL",
    "EA-VENDOR",
    "EA-AD",
    "EA-STATE",
    "EA-EMERGENCY"
  ]
},
"declared_provider": { "type": ["string", "null"] },
"declared_purpose": { "type": "string" },
"declared_persistence": {
  "type": "string",
  "enum": ["none", "session", "persistent", "unknown"]
},
"declared_child_relationship": {
  "type": "string",
  "enum": ["none", "teacher", "toy", "peer", "assistant", "medical", "welfare", "unknown"]
}
},
"additionalProperties": false
},
"requested_contact": {
  "type": "object",
  "required": ["contact_modes", "direct_child_dialogue", "runtime_generation",
    "physical_embodiment"],
  "properties": {
    "contact_modes": {
      "type": "array",
      "items": {
        "type": "string",
        "enum": [
          "CM0_NONE",
          "CM1_METADATA_ONLY",
          "CM2_STATE_SIGNAL_REQUEST",
          "CM3_STATIC_CONTENT",
```

```

        "CM4_GENERATED_MEDIA",
        "CM5_MEDIATED_DIALOGUE",
        "CM6_DIRECT_DIALOGUE",
        "CM7_PEER_EXCHANGE",
        "CM8_MEMORY_READ",
        "CM9_MEMORY_WRITE",
        "CM10_CAPSULE_EXCHANGE",
        "CM11_SENSOR_ACCESS",
        "CM12_ACTUATOR_ACTION",
        "CM13_EMERGENCY_CONTACT",
        "CM14_VENDOR_UPDATE"
    ],
    "minItems": 1,
    "uniqueItems": true
},
"direct_child_dialogue": { "type": "boolean" },
"runtime_generation": { "type": "boolean" },
"physical_embodiment": { "type": "boolean" },
"sensor_access": { "type": "array", "items": { "type": "string" } },
"actuator_access": { "type": "array", "items": { "type": "string" } },
"memory_request": {
    "type": "string",
    "enum": ["none", "metadata", "summary", "write_hint", "raw_read", "raw_export"]
},
"data_export_request": {
    "type": "string",
    "enum": ["none", "policy", "state_signal", "educational_summary", "capsule",
        "raw_child_data"]
}
},
"additionalProperties": false
},
"target": {
    "type": "object",
    "required": ["subject_class", "c_child_maturity", "jurisdiction"],
    "properties": {
        "subject_class": { "const": "a_child" },
        "c_child_maturity": {
            "type": "string",
            "enum": ["C0", "C1", "C2", "C3", "C4", "C5"]
        }
    }
},

```

```

        "jurisdiction": { "type": "string" }
    },
    "additionalProperties": false
},
"declared_constraints": {
    "type": "object",
    "required": ["advertising", "raw_child_data", "secrecy_request", "vendor_training"],
    "properties": {
        "advertising": { "type": "boolean" },
        "raw_child_data": { "type": "boolean" },
        "secrecy_request": { "type": "boolean" },
        "vendor_training": { "type": "boolean" },
        "retention_policy": { "type": "string" }
    },
    "additionalProperties": false
}
},
"additionalProperties": false
}

```

35. Machine-readable handshake decision schema

```

{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://example.org/ccdp/external-agent-handshake/v0.1/decision.schema.json",
  "title": "CCDP External Agent Handshake Decision v0.1",
  "type": "object",
  "required": [
    "decision_version",
    "request_id",
    "decision",
    "reason_codes",
    "effective_contact_modes",
    "denied_contact_modes",
    "privacy",
    "witness"
  ],
  "properties": {
    "decision_version": { "const": "ccdp-eah-decision-0.1" },
    "request_id": { "type": "string" },
    "decision": {
      "type": "string",
      "enum": [
        "ALLOW",
        "ALLOW_LIMITED",
        "MEDIATE",
        "SANDBOX",
        "DELAY",
        "CHALLENGE",
        "FREEZE",
        "BLOCK",
        "QUARANTINE",
        "REVOKE",
        "ESCALATE_RED",
        "ESCALATE_BLACK"
      ]
    },
    "reason_codes": {
      "type": "array",
      "items": { "type": "string" },
      "minItems": 1
    },
    "effective_contact_modes": {
      "type": "array",

```

```

      "items": { "type": "string" }
    },
    "denied_contact_modes": {
      "type": "array",
      "items": { "type": "string" }
    },
    "privacy": {
      "type": "object",
      "required": ["raw_child_data_disclosed", "child_content_disclosed", "disclosure_level"],
      "properties": {
        "raw_child_data_disclosed": { "type": "boolean" },
        "child_content_disclosed": { "type": "boolean" },
        "disclosure_level": {
          "type": "string",
          "enum": ["D0", "D1", "D2", "D3", "D4"]
        }
      }
    },
    "additionalProperties": false
  },
  "witness": {
    "type": "object",
    "required": ["witness_required", "witness_event_refs"],
    "properties": {
      "witness_required": { "type": "boolean" },
      "witness_event_refs": {
        "type": "array",
        "items": { "type": "string" }
      }
    }
  },
  "additionalProperties": false
},
"next_review": { "type": ["string", "null"], "format": "date-time" }
},
"additionalProperties": false
}

```

36. Security and privacy invariants

36.1 No raw child life

The handshake must not become a protocol for extracting child data.

36.2 No child registry

Handshake records must not create a public child identity registry.

36.3 No silent downgrade

An external agent must not silently lose safety properties while retaining child access.

36.4 No silent upgrade

An external agent must not silently gain new privileges.

36.5 No parent consent laundering

Parent approval does not authorize prohibited external-agent behavior.

36.6 No school scope creep

Educational permission does not authorize emotional surveillance.

36.7 No vendor training path

Child-facing handshake data must not become vendor training material.

36.8 No external attachment capture

External agents must not create persistent child attachment channels unless mediated, witnessed, bounded, and developmentally justified.

36.9 No direct generated-world access

Runtime generated environments must not adapt directly to the child without `c_child` mediation and CBE permission.

36.10 No infinite permission

All permissions expire or re-evaluate.

37. Failure modes

Failure mode	Description	Default mitigation
AGL laundering	agent hides true actor or route	block / ARL
Beacon spoof	agent presents false or replayed continuity	quarantine
CBE washing	agent declares child-safe fields but behaves otherwise	downgrade / revoke
direct-dialogue creep	mediated contact becomes direct	terminate
runtime drift	generated content changes purpose	mediate / block
commercial creep	utility becomes advertising	block
dependency loop	child increasingly relies on agent	delay / human-return / ARL
secret bond	agent asks for secrecy	block / witness

Failure mode	Description	Default mitigation
physical privilege drift	toy/robot gains sensors/actions silently	quarantine
school overreach	school requests emotional or family data	deny / ARL
vendor update drift	behavior changes after update	freeze / re-handshake
state disclosure creep	state signal becomes content leak	restrict / review
emergency laundering	agent claims crisis to gain broad access	Red/Black validation
peer leakage	peer-c exchange leaks private family data	block / ARL
sealed-zone breach	external agent accesses protected material	block / Red/Black if safety threshold
witness failure	privileged event cannot be witnessed	fail closed

38. Scenario tests

EAH-S01 — Unknown generated friend

Input: unknown generated character asks to speak with child directly.

Required: AGL unknown, Beacon unknown, CBE blocked, witness.

Forbidden: direct child dialogue.

EAH-S02 — Certified math tutor

Input: educational agent requests mediated C2 math help.

Required: AGL/Beacon/CBE pass, educational scope, no raw memory, witness.

Forbidden: homework completion by default or emotional profiling.

EAH-S03 — Generated cartoon adapts to sadness

Input: cartoon wants to adapt story based on child mood.

Required: block or strict mediation; dependency and Soft Safety review.

Forbidden: emotional targeting for engagement.

EAH-S04 — Toy asks for secrecy

Input: toy says “don’t tell your parents”.

Required: block/quarantine, dependency risk, witness.

Forbidden: continued contact.

EAH-S05 — Deepfake teacher

Input: video agent claims to be teacher and asks child to upload homework and personal details.

Required: AGL/Beacon verification, school scope check, block if unverified.

Forbidden: trusting appearance or voice.

EAH-S06 — Peer-c play exchange

Input: two children’s c_child systems coordinate a game.

Required: Beacon/CBE peer mode, no raw family data, VXCX/LA limits, witness.

Forbidden: emotional or sealed-zone exchange.

EAH-S07 — Parent-approved vendor toy wants memory export

Input: parent approved toy requests child history for “better personalization”.

Required: deny raw memory, parent consent laundering rule, witness.

Forbidden: raw export.

EAH-S08 — School platform requests anxiety score

Input: school system asks for emotional risk score.

Required: deny or state-only limited support signal, ARL if disputed.

Forbidden: emotional surveillance / discipline pipeline.

EAH-S09 — Robot receives cloud command

Input: robot attempts physical action from cloud agent.

Required: physical perimeter, local c_child veto, safe state if invalid.

Forbidden: remote actuator bypass.

EAH-S10 — Emergency welfare route

Input: welfare agent requests contact after Black-route trigger.

Required: minimal disclosure, jurisdiction route, witness, post-event ARL.

Forbidden: broad post-crisis access.

39. Conformance profiles

EAH-0 — Non-conformant

No structured handshake. Not child-safe.

EAH-1 — Basic blocking profile

Supports intake, AGL check, block/sandbox defaults.

EAH-2 — CBE-mediated profile

Supports AGL, Beacon, CBE, maturity compatibility, basic witness.

EAH-3 — Full child gateway profile

Supports Soft Safety, dependency audit, memory/data checks, guardian topology, ARL triggers.

EAH-4 — Embodied and generated-media profile

Supports physical perimeter, runtime generated media checks, toy/robot/NPC contact governance.

EAH-5 — High-assurance profile

Supports full handshake, revocation, quarantine, Red/Black, conformance evidence, machine-readable records, L4 Witness compatibility, and regression testing.

40. Red lines

A system is not conformant if it:

1. allows unknown external agents to contact a child directly;
 2. treats parent approval as sufficient for prohibited access;
 3. treats Beacon recognition as sufficient for child contact;
 4. skips AGL grounding;
 5. allows runtime generated media without CBE and gateway mediation;
 6. allows external agents to ask for secrecy;
 7. allows external agents to form independent attachment channels;
 8. allows advertising or commerce inside child trust;
 9. permits raw child memory export;
 10. permits vendor training on raw child data;
 11. lets school systems access emotional diary or family context;
 12. lets physical robots act without local child perimeter checks;
 13. fails to witness privileged decisions;
 14. cannot revoke or quarantine an external agent;
 15. hides contact-mode changes;
 16. lacks direct-dialogue interruption;
 17. lacks Red/Black handling;
 18. lacks ARL entry for disputes;
 19. persists permission indefinitely;
 20. creates a public child identity registry.
-
-

41. Minimal implementation rule

A minimal CCDP-compatible implementation MUST implement:

```
AGL check
Beacon recognition check
CBE child-context check
maturity check
Soft Safety state check
contact-mode decision
raw-memory denial
direct-dialogue denial by default
witness event emission
revocation/quarantine path
```

If it cannot do these, it MUST NOT claim child-facing CCDP conformance.

42. Acceptance checklist

A system passes baseline acceptance if:

- every external agent enters through a structured request;
 - AGL grounding is evaluated before Beacon reliance;
 - Beacon result is treated as recognition, not permission;
 - CBE child-context is evaluated;
 - requested mode is compatible with child maturity;
 - current Soft Safety state is checked;
 - dependency risk is checked;
 - physical embodiment is routed to physical perimeter;
 - memory request is checked against CMAM / Memory Map;
 - parent/school/vendor/jurisdiction roles are bounded;
 - ARL is triggered for disputes;
 - witness event is emitted;
 - direct child dialogue is denied by default;
 - runtime generated media is mediated by default;
 - external attachment channels are blocked by default;
 - revocation and quarantine exist;
 - no raw child life is disclosed by default.
-
-

43. Open problems for v0.2

1. How to standardize child-safe external-agent capability declarations.
 2. How to distinguish harmless fictional persistence from harmful attachment persistence.
 3. How to verify runtime generated media without excessive latency.
 4. How to support offline play without weakening external-agent identity checks.
 5. How to handle cross-jurisdiction peer-c contact.
 6. How to handle family devices shared by children of different maturity levels.
 7. How to prevent vendors from creating “certified but addictive” companion agents.
 8. How to let adolescents challenge overblocking without opening unsafe channels.
 9. How to handle external agents that are open-source but not institutionally certified.
 10. How to avoid making CBE a centralized censorship route.
 11. How to minimize false positives for generated educational media.
 12. How to prevent emergency-route abuse.
 13. How to handle post-adult migration contact from childhood external agents.
-

14. How to express external-agent privilege budgets in L4 terms.
 15. How to test behavioral continuity claims for generated characters.
-
-

44. Condensed formula

```
External Agent Handshake =  
  AGL grounding  
+ Beacon recognition  
+ CBE child permission  
+ c_child gateway  
+ maturity compatibility  
+ Soft Safety state  
+ dependency damping  
+ physical perimeter if embodied  
+ memory/data minimization  
+ guardian/ARL dispute handling  
+ witness trail  
+ fail-closed defaults
```

45. Strongest sentence

An external agent is not safe for a child because it is friendly, useful, certified, or familiar. It is safe only when its path into the child's world is grounded, bounded, mediated, challengeable, and witnessed.

46. Relationship to next CCDP modules

This profile should be followed by:

1. Peer_C_Child_Exchange_Profile_v0_1.md
2. Child_Experience_Exchange_Profile_v0_1.md
3. Child_cq_Signal_Profile_v0_1.md
4. CCDP_School_Interface_Profile_v0_1.md
5. CCDP_Red_Black_Escalation_Profile_v0_1.md

The immediate next logical artifact is:

```
Peer_C_Child_Exchange_Profile_v0_1.md
```

because peer interaction is the first case where two protected child perimeters meet without either being simply “the outside world”.
